

knowledge-checks\knowledge-check-sep-13.py

```
1 '''
2 next problem
3 '''
4  $(-4)^{-1} =$ 
5  $(1 / (-4)) =$ 
6  $[- (1 / 4)]$ 
7
8 '''
9 next problem
10 '''
11  $562 = b + cb$ 
12  $562 = b + (62 + b)$ 
13
14  $562 = 2b + 62$ 
15  $- 62 \quad - 62$ 
16 #-----
17  $500 = 2b$ 
18  $/ 2 \quad / 2$ 
19 #-----
20  $[250 = b]$ 
21
22 '''
23 next problem
24 '''
25 #find f(4) and g(-2)
26
27  $f(x) = (-2x) + (-1)$ 
28  $f(4) = (-2(4)) + (-1)$ 
29  $= (-8) + (-1)$ 
30  $[f(4) = (-9)]$ 
31
32
33  $g(x) = (-3)x^3 + (-2)$ 
34  $g(-2) = (-3)(-2)^3 + (-2)$ 
35  $= (-3)(-8) + (-2)$ 
36  $[g(-2) = 22]$ 
37
38 '''
39 next problem
40 '''
41  $((-9), (-2))$ 
42
43  $5(-9) + (-8)(-2) = (-29)$ 
44  $(-45) + 16 = (-29)$ 
45  $(-29) = (-29)$ 
46 [yes]
47
48  $((-1), 3)$ 
```

```
49
50  $5(-1) + (-8)(3) = (-29)$ 
51  $(-5) + (-24) = (-29)$ 
52  $(-29) = (-29)$ 
53 [yes]
54
55  $(0, (-4))$ 
56
57  $0 + (-8)(-4) = (-29)$ 
58  $(-12) \neq (-29)$ 
59 [no]
60
61  $(6, 5)$ 
62
63  $5(6) + (-8)(5) = (-29)$ 
64  $30 + (-40) = (-29)$ 
65  $(-10) \neq (-29)$ 
66 [no]
67
68 '''
69 next problem
70 '''
71  $4x + 2y = 18$ 
72  $y = ((-2x) + 9)$ 
73
74  $4x + (2)((-2x) + 9) = 18$ 
75  $4x + (-4x) + 18 = 18$ 
76  $18 = 18$ 
77
78 [they are the same line]
79
80 '''
81 slope-intercept form →
82  $y = mx + b$ 
83 '''
84
85  $4x + 2y = 18$ 
86  $- 4x \quad - 4x$ 
87 #-----
88  $2y = (-4x) + 18$ 
89  $/ 2 \quad / 2 \quad / 2$ 
90 #-----
91  $(y = (-2x) + 9)$ 
92 and
93  $(y = (-2x) + 9)$ 
94 [are identical]
95
96  $y = (-2x) + 9$ 
97
98  $19 = (-2(-5)) + 9$ 
```

```

99      19 = 10 + 9
100     [yes]
101
102     y = (-2x) + 9
103
104     1 = (-2(4)) + 9
105     1 = (-8) + 9
106     [yes]
107
108     y = (-2x) + 9
109     0 = (-2(6)) + 9
110     0 = (-12) + 9
111     0 ≠ (-3)
112     [no]
113
114     y = (-2x) + 9
115
116     (-3) = (-2(5)) + 9
117     (-3) = (-10) + 9
118     (-3) ≠ (-1)
119     [no]
120
121     '''
122     next problem
123     '''
124     (v)2 - 9
125
126     '''
127     difference of squares →
128     a2 - b2 = (a - b)(a + b)
129     '''
130     (v)2 - 9 =
131     (v)2 - (3)2 =
132     a = v
133     b = 3
134     (v - 3)(v + 3)
135
136     '''
137     next problem
138     '''
139     '''
140     pythagorean theorem →
141     a2 + b2 = c2
142     right triangles → c = hypotenuse (longest side across from 90° angle)
143     '''
144
145     c = 16
146     a = 8
147     b = x
148

```

```

149     82 + x2 = 162
150     64 + x2 = 256
151     - 64      - 64
152     #-----
153     x2 = 192
154     √x2 = √192
155     x = 13.856
156     [x = 13.86]
157
158     '''
159     next problem
160     '''
161     '''
162     slope → rise over run →
163     m = (y2 - y1)
164           -----
165           (x2 - x1)
166     '''
167
168     (2, (-2)) → (point-1)
169
170     ((-3), 2) → (point-2)
171
172     m = (2 - (-2))
173     #-----
174     ((-3) - 2)
175
176     (2 + 2)
177     #-----
178     (-5)
179
180     4
181     #-----
182     (-5)
183
184     [m = -(4 / 5)]
185
186     '''
187     next problem
188     '''
189     (1 / 2x) + (4 / 9x)
190
191     '''
192     least common denominator
193     '''
194
195     (1 / 2x) + (4 / 9x)
196     * 9      * 2
197     #-----
198

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```

199      (9 / 18x) + (8 / 18x) =
200      [17 / 18x]
201
202      '''
203      next problem
204      '''
205      (-2)(5x + (-5)) + 6x = 6(x + 2)
206      (-10x) + 10 + 6x = 6x + 12
207
208      (-4x) + 10 = 6x + 12
209      + 4x - 12 + 4x - 12
210      #-----
211
212      (-2) = 10x
213      / 10 / 10
214      #-----
215      [- (1 / 5) = x]
216
217      '''
218      next problem
219      '''
220      5x + (-4y) = 24
221      x = (3y + 7)
222
223      5(3y + 7) + (-4y) = 24
224      15y + 35 + (-4y) = 24
225
226      11y + 35 = 24
227      (-35) (-35)
228      #-----
229      11y = (-11)
230      / 11 / 11
231      #-----
232      [y = (-1)]
233
234      x = (3(-1) + 7)
235      x = (-3) + 7
236      [x = 4]
237
238
239      '''
240      next problem
241      '''
242      f(5) and g(-2)
243
244      f(x) = (-5x) + 4
245      g(x) = 2x^2 + (-3x)
246
247      f(5) = (-5)(5) + 4
248      = (-25) + 4

```

```

249      = (-21)
250      [f(5) = (-21)]
251
252      g(-2) = 2(-2)^2 + (-3)(-2)
253      = 2(4) + 6
254      = 8 + 6
255      = 14
256      [g(-2) = 14]
257
258
259      '''
260      next problem
261      '''
262      '''
263      midpoint → average of x, average of y →
264      M = ( (x1 + x2) / 2 , (y1 + y2) / 2 )
265      ---
266      2 2
267      '''
268      point-1 = ((-7),7)
269      point-2 = (5, (-1))
270
271      M = ((-7) + 5) / 2 , (7, (-1)) / 2
272      #-----
273      2 2
274
275      M = ((-2) / 2 , 6 / 2)
276      [M = (-1, 3)]
277
278
279      '''
280      next problem
281      '''
282      '''
283      circle → circumference (distance around) →
284      C = 2 * π * r or C = π * d
285
286      circle → area (space inside) →
287      A = π * r^2
288      '''
289      units = (cm)
290      r = 9
291
292      C = 2(3.14)(9)
293      [C = 56.52 cm]
294
295      A = (3.14)(9)^2
296      [A = 254.34 cm^2]
297
298

```

```

299 '''
300 next problem
301 '''
302 '''
303 area of triangle with base and height →
304     A = 0.5 * b * h
305 '''
306     A = (0.5)(18)(5)
307     [A = 45 cm²]
308
309
310 '''
311 next problem
312 '''
313     (4x + 7)(6x + (-3u) + (-5))
314
315
316         [6x]      [(-3u)]      [(-5)]
317     [4x]    24x²    (-12ux)    (-20x)
318     [7]     42x     (-21u)     (-35)
319
320     24x² + (-12ux) + [(-20x) + 42x] + (-21u) + (-35) =
321     [(-20x) + 42x] = [[22x]]
322
323     [24x² + (-12ux) + [[22x]] + (-21u) + (-35)]
324
325
326 '''
327 next problem
328 '''
329     (-5)(-3) + (1 + (-2³))² =
330     15 + (1 + (-8))² =
331     15 + (-7)² =
332     15 + 49 =
333     [64]
334
335
336
337
338
339

```